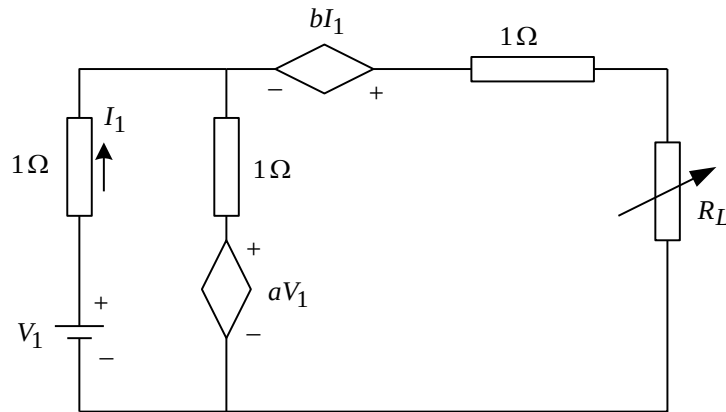
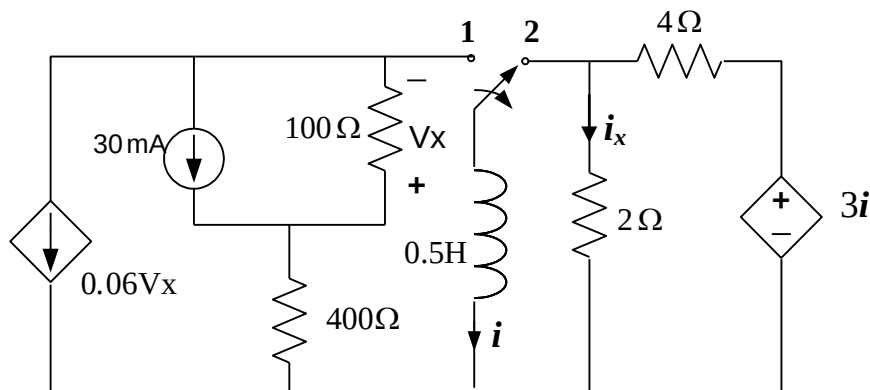


Answer all questions

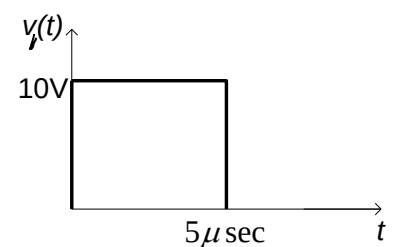
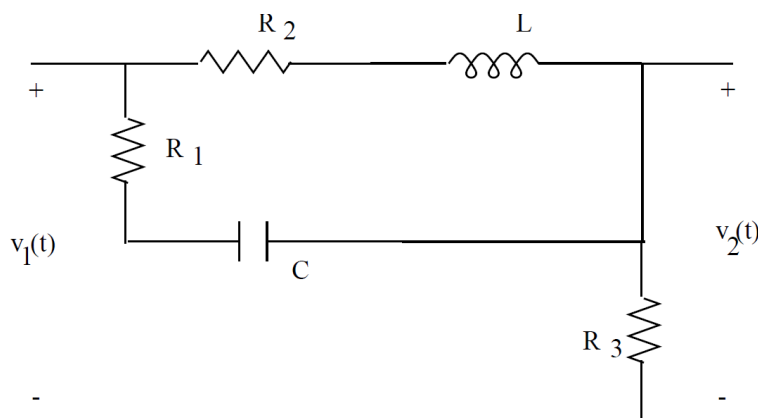
1. In the circuit shown in the figure, $V_1 = 10\text{ V}$, $a = 1.5$ and $b = 2$. Determine the value of R_L for maximum power transfer. Also determine the maximum value of power. [6]



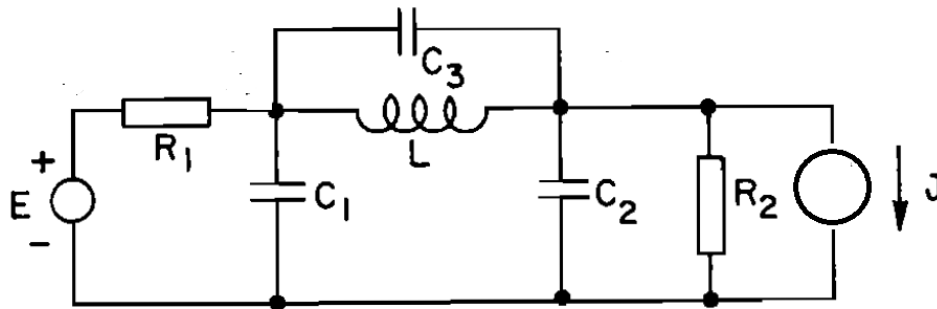
2. In the circuit in figure, the switch was in position 1 for a long time. At time $t = 0$ the switch is moved to position 2. Find i_x , the current through 2Ω as a function of time. [10]



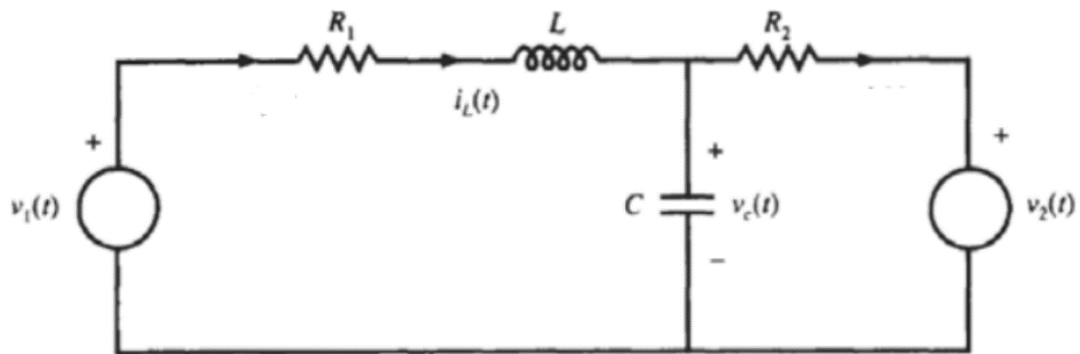
3. In the circuit shown $L = RC^2$ and $R_1 = R_2$. If $v_1(t)$ is a pulse as shown in figure, determine $v_2(t)$ using suitable network function. [5]



4. Transform the current source into equivalent voltage source, draw appropriate graph and obtain graph theoretic formulation of loop equations. Take all the resistance branches as tree branches. [7]



5. Find a state space model for the circuit shown in figure. The outputs are the voltages across R_1 , R_2 and L [4]



6. What is the output of the following MATLAB code? [8]

```
ts=[0 2];
x0=[1;0];
[t,x]=ode45('midterm',ts,x0);
a=[t,x];
[m,n]=size(a);
a(m,2)
```

```
function c = midterm(t,x)
c=[x(2);-2*x(1)-3*x(2)+1];
```